Yonwoo Choi

Algorithm Engineer
GenGenAI

Email: yonwoo@gengen.ai, **Web**: yc4ny.github.io, **Google Scholar**: Link

EDUCATION

Seoul National University *Feb 2024*
M.S. in Computer Science and Engineering
Advisor: Prof. Hanbyul Joo
Thesis: A Wrist Mounted Camera System for Hand Pose Estimation from Single RGB Images

University of Virginia *May 2021*
B.A. in Computer Science

RESEARCH INTEREST

My research lies in 3D computer vision centered on humans, with a focus on capturing, understanding, and reconstructing people in 3D. Ultimately, I aim to build machines that perceive and understand human motion and behavior as naturally as people understand one another. I pursue this direction by building practical capture systems and learning-based methods that bring humans into the digital world.

PUBLICATIONS

"SVAD: From Single Image to 3D Avatar via Synthetic Data Generation with Video Diffusion and Data Augmentation"
Yonwoo Choi
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) Workshop, 2025.

"A Wrist Mounted Camera System for Hand Pose Estimation from Single RGB Images"
Yonwoo Choi, Hanbyul Joo
M.S. Thesis, Seoul National University, 2024.

WORK EXPERIENCE

GenGenAI, South Korea
Algorithm Engineer (Alt. Military Service) *Sept. 2025 – Present*
Worked on Controllable Image and Video Generation Across Domains

SECERN AI, South Korea
Research Engineer (Alt. Military Service) *Mar. 2024 – Sept. 2025*
Worked on Animatable 3D Human Avatars

OPEN SOURCE

SVAD: `https://github.com/yc4ny/SVAD`
Official implementation of the paper SVAD

Facebuilder: `https://github.com/yc4ny/facebuilder`
Multiview face modeling with real-time 3D consistency

SMPL Fitting: `https://github.com/yc4ny/SMPL-Fitting`
Parametric human shape and pose optimization

Egocentric Human Pose Estimator: `https://github.com/yc4ny/xR-EgoPose`
Egocentric 3D human pose estimation from a single fisheye camera

Multiview Human Pose Capture System
`https://github.com/yc4ny/Multi-Camera-3D-Human-Reconstruction`
5 camera system setup for 3D human reconstruction

PROJECTS

Missile-Perspective Naval Scene Video Generation *Dec. 2025 – Present*
Missile-perspective synthetic video generation for naval target detection
GenGenAI, Hanwha Aerospace

Missile-Perspective Ground Scene Video Generation *Mar. 2026 – July. 2026*
Missile-perspective synthetic video generation for ground vehicle detection
GenGenAI, Hanwha Aerospace

Multiview 3D X-Ray Detection for Airport Screening *Mar. 2024 – Sept. 2025*
3D X-ray calibration, 3D bounding box detection
SECERN AI, Ministry of Land, Infrastructure and Transport

TEACHING

Teaching Assistant, Seoul National University *Fall 2023*
4190.408 Artificial Intelligence (Instructor: Hanbyul Joo)

Teaching Assistant, Seoul National University *Spring 2022*
4190.773 3D Computer Vision (Instructor: Hanbyul Joo)